Analysis-0 2026 Spring

Homework 6

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*All of my homework is available at `https://github.com/lacampanella233/Homework.git`.

Problem 1 (3-8). If $\sum a_n$ converges and $\{b_n\}$ is monotone and bounded, prove that

$$\sum a_n b_n$$

also converges.

Proof. Denote $s_n = a_0 + a_1 + \cdots + a_n$. By the Abel summation formula, we have

$$S_n := \sum_{k=1}^{n} a_k b_k = s_n b_n + \sum_{k=1}^{n-1} s_k(b_k - b_{k+1}).$$

We will prove that $\sum a_n b_n$ is a Cauchy series, hence $\sum a_n b_n$ converges. Note that $\{s_n\}$ and (b_n) all converge, thus $\{s_n b_n\}$ also converges, and there exist $M > 0$ such that $|s_n| < M$ for all $n \in \mathbb{N}_+$.

For any positive number ε, there exist $N_1, N_2 \in \mathbb{N}_+$ such that

$$|s_m b_m - s_n b_n| < \frac{\varepsilon}{2}, \forall\, m > n > N_1; \quad |b_m - b_n| < \frac{\varepsilon}{2M}, \forall\, m > n > N_2.$$

Let $N = \max\{N_1, N_2\}$. For any $m > n > N$, then for all $m > n > N$,

$$\begin{aligned}
|S_m - S_n| &= \left| s_m b_m - s_n b_n + \sum_{k=n}^{m-1} s_k(b_k - b_{k+1}) \right| \\
&\leq |s_m b_m - s_n b_n| + \sum_{k=n}^{m-1} |s_k| \cdot |b_k - b_{k+1}| < \frac{\varepsilon}{2} + M \sum_{k=n}^{m-1} |b_k - b_{k+1}| \\
&= \frac{\varepsilon}{2} + M\,|b_n - b_m| \text{ (This is because } \{b_n\} \text{ is monotone)} \\
&< \varepsilon.
\end{aligned}$$

Therefore, $\sum a_n b_n$ is a Cauchy series, and $\sum a_n b_n$ converges. $\square$

Problem 2 (3-9). Find the radius of convergence of each of the following power series: (a) $\sum n^3 z^n$, (b) $\sum \frac{2^n}{n!} z^n$, (c) $\sum \frac{2^n}{n^2} z^n$, (d) $\sum \frac{n^3}{3^n} z^n$.

Solution. (a)

$$R = \frac{1}{\limsup \sqrt[n]{n^3}} = \frac{1}{1} = 1.$$

(b)

$$R = \frac{1}{\limsup \sqrt[n]{2^n/n!}} = \frac{1}{0} = \infty.$$

(c)

$$R = \frac{1}{\limsup \sqrt[n]{2^n/n^2}} = \frac{1}{2\limsup n^{-2/n}} = \frac{1}{2}.$$

(d)

$$R = \frac{1}{\limsup \sqrt[n]{n^3/3^n}} = \frac{1}{3\limsup n^{-3/n}} = \frac{1}{3}.$$

Problem 3 (3-10). Assume that all coefficients of the power series

$$\sum a_n z^n$$

are integers, and infinitely many of them are nonzero. Prove that its radius of convergence is at most 1.

Proof. It suffices to show that $\limsup_{n\to\infty} \sqrt[n]{|a_n|} \geq 1$. For any positive integer N, there exists $n > N$ such that $|a_n| \geq 1$ (since infinitely many of the coeffecients are nonzero), then the conclusion holds. □

Problem 4 (3-14). Let $\{s_n\}$ be a sequence of complex numbers, and define its arithmetic means by

$$\sigma_n = \frac{s_0 + s_1 + \cdots + s_n}{n+1}, \qquad (n = 0, 1, 2, \ldots).$$

(a) If $\lim_{n\to\infty} s_n = s$, prove that $\lim_{n\to\infty} \sigma_n = s$.

(b) Construct a sequence $\{s_n\}$ such that $\lim_{n\to\infty} \sigma_n = 0$ but $\{s_n\}$ does not converge.

(c) Can the following happen: $s_n > 0$ for all n, $\lim_{n\to\infty} \sigma_n = 0$, but $\limsup_{n\to\infty} s_n = \infty$?

(d) For $n \geq 1$, let $a_n = s_n - s_{n-1}$. Prove that

$$s_n - \sigma_n = \frac{1}{n+1} \sum_{k=1}^{n} k a_k.$$

Assume further that $\lim_{n\to\infty}(na_n) = 0$ and that $\{\sigma_n\}$ converges. Prove that $\{s_n\}$ converges. (This is a partial converse of (a), with the additional assumption $na_n \to 0$.)

(e) Derive the conclusion in (d) under weaker hypotheses: assume $M < \infty$, $|na_n| \leq M$ for all n, and $\lim_{n\to\infty} \sigma_n = \sigma$. Prove $\lim_{n\to\infty} s_n = \sigma$ by following these steps:

If $m < n$, then

$$s_n - \sigma_n = \frac{m+1}{n-m}(\sigma_n - \sigma_m) + \frac{1}{n-m} \sum_{i=m+1}^{n} (s_n - s_i).$$

For such i,

$$|s_n - s_i| \leq \frac{(n-i)M}{i+1} \leq \frac{(n-m-1)M}{m+2}.$$

Fix $\varepsilon > 0$. For each n, choose a positive integer m such that

$$m \leq \frac{n-\varepsilon}{1+\varepsilon} < m+1.$$

Then

$$\frac{m+1}{n-m} \leq \frac{1}{\varepsilon},$$

and $|s_n - s_i| < M\varepsilon$. Hence

$$\limsup_{n\to\infty} |s_n - \sigma| \leq M\varepsilon.$$

Since ε is arbitrary, conclude that $\lim_{n\to\infty} s_n = \sigma$.

Proof. (a) For any positive number ε, there exists $N_1 \in \mathbb{N}_+$ such that $|s_n - s| < \varepsilon/3$ for all $n > N_1$. Take $N_2 \in \mathbb{N}_+$ such that $s_0 + s_1 + \cdots + s_{N_1} < (\varepsilon/3)N_2$. It's clear that there exists $N_3 \in \mathbb{N}_+$ such that $(N_1+1)s < N_3\varepsilon/3$.

Set $N = \max\{N_1, N_2, N_3\}$. For any $n > N$,

$$\begin{aligned}|\sigma_n - s| &\leq \left|\frac{s_0 + s_1 + \cdots + s_{N_1}}{n}\right| + \left|\frac{s_{N_1+1} + s_{N_1+2} + \cdots + s_n}{n} - s\right| \\ &\leq \frac{\varepsilon}{3} + \left|\frac{(n-N_1)s}{n} - s\right| + \frac{|s_{N_1+1} - s| + |s_{N_1+2} - s| + \cdots + |s_n - s|}{n} \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.\end{aligned}$$

Hence $\lim_{n\to\infty} \sigma_n = s$.

(b) Let $s_n = (-1)^n$. Then $\sigma_n = \frac{1}{n+1}\sum_{k=0}^{n}(-1)^k$. It's easy to verify that $\lim_{n\to\infty}\sigma_n = 0$, but $\{s_n\}$ does not converge.

(c) Yes. Let

$$s_n = \begin{cases} k\,, & \text{if } n = 2^k; \\ 2^{-n}, & \text{otherwise.}\end{cases}$$

It's not hard to verify that $s_n = \mathcal{O}(\log n)$, then $\lim_{n\to\infty}\sigma_n = 0$. But by the definition of s_n, we have $\limsup_{n\to\infty} s_n = +\infty$.

(d) Note that

$$\begin{aligned}\sum_{k=1}^{n} ka_k = \sum_{k=1}^{n} k(s_k - s_{k-1}) = \sum_{k=1}^{n} ks_k - \sum_{k=1}^{n} ks_{k-1} = ns_n + \sum_{k=1}^{n-1} ks_k - \sum_{k=0}^{n-1}(k+1)s_k \\ = ns_n - \sum_{k=0}^{n-1} s_k = (n+1)(s_n - \sigma_n).\end{aligned}$$

Suppose $\lim_{n\to\infty} na_n = 0$ and $\{\sigma_n\}$ converges. We only need to prove that $\lim_{n\to\infty}(s_n - \sigma_n) = 0$. This is an immediate consequence of (a), with subsituting s_n to na_n.

(e) Assume $M < \infty$, $|na_n| \leq M$ for all n, and $\sigma_n \to \sigma$. For $m < n$, the identity in the statement gives

$$s_n - \sigma_n = \frac{m+1}{n-m}(\sigma_n - \sigma_m) + \frac{1}{n-m}\sum_{i=m+1}^{n}(s_n - s_i).$$

For each $i \in \{m+1, \dots, n\}$, $s_n - s_i = \sum_{k=i+1}^{n} a_k$, hence

$$|s_n - s_i| \le \sum_{k=i+1}^{n} |a_k| \le \sum_{k=i+1}^{n} \frac{M}{k} \le \frac{(n-i)M}{i+1} \le \frac{(n-m-1)M}{m+2}.$$

Fix $\varepsilon > 0$. For each n, choose $m \in \mathbb{N}_+$ such that

$$m \le \frac{n-\varepsilon}{1+\varepsilon} < m+1.$$

Then $n - m \ge \varepsilon(m+1) + \varepsilon > \varepsilon(m+1)$, so

$$\frac{m+1}{n-m} \le \frac{1}{\varepsilon}.$$

Also, $n - m - 1 < \varepsilon(m+2)$, therefore for $i = m+1, \dots, n$,

$$|s_n - s_i| \le \frac{(n-m-1)M}{m+2} < M\varepsilon.$$

Taking absolute values in the decomposition of $s_n - \sigma_n$, we get

$$|s_n - \sigma_n| \le \frac{m+1}{n-m} |\sigma_n - \sigma_m| + \frac{1}{n-m} \sum_{i=m+1}^{n} |s_n - s_i| \le \frac{1}{\varepsilon} |\sigma_n - \sigma_m| + M\varepsilon.$$

Since $m \to \infty$ as $n \to \infty$ and $\sigma_n \to \sigma$, we have $|\sigma_n - \sigma_m| \to 0$, hence $\limsup_{n\to\infty} |s_n - \sigma_n| \le M\varepsilon$. Finally, $|s_n - \sigma| \le |s_n - \sigma_n| + |\sigma_n - \sigma|$, and $\sigma_n \to \sigma$ implies

$$\limsup_{n\to\infty} |s_n - \sigma| \le M\varepsilon.$$

Because $\varepsilon > 0$ is arbitrary, $\lim_{n\to\infty} s_n = \sigma$. $\square$

Problem 5 (4-1). Let f be a real-valued function defined on $\mathbb{R}^1$. Assume that for every $x \in \mathbb{R}^1$,

$$\lim_{h\to 0} \big[f(x+h) - f(x-h)\big] = 0.$$

Does this imply that f is continuous?

Solution. No. Let

$$f(x) = \begin{cases} 1, & \text{if } x = 0; \\ 0, & \text{if } x \ne 0. \end{cases}$$

Then for any $x \in \mathbb{R}^1$,

$$\lim_{h\to 0} \big[f(x+h) - f(x-h)\big] = 0,$$

but f is not continuous at $x = 0$.

Problem 6 (4-3). Let f be a continuous real-valued function on a metric space X. Let

$$Z(f) = \{p \in X : f(p) = 0\}$$

(the zero set of f). Prove that $Z(f)$ is closed.

Proof. We will prove that for any $p \in X$ such that there exists $\{p_n\}_{n\geq 1} \subset Z(f)$ such that $\lim_{n\to\infty} p_n = p$, then $p \in Z(p)$.

Suppose the contray, then $f(p) \neq 0$. Since f is continuous at p, there exists $\varepsilon > 0$ such that for any $q \in X$ with $d(p,q) < \varepsilon$, we have $|f(q) - f(p)| < |f(p)|/2$. In particular, for sufficiently large n, we have $d(p, p_n) < \varepsilon$, hence

$$|f(p_n) - f(p)| < \frac{|f(p)|}{2}.$$

However, $f(p_n) = 0$ for all $n \in \mathbb{N}_+$, so $|f(p)| < |f(p)|/2$, which is a contradiction. Therefore, $p \in Z(f)$, and $Z(f)$ is closed. □

Problem 7 (4-7). If $E \subset X$ and f is a function defined on X, then the restriction of f to E means the function g with domain E such that for every $p \in E$, $g(p) = f(p)$.

Define f and g on $\mathbb{R}^2$ by

$$f(0,0) = g(0,0) = 0,$$

and for $(x,y) \neq (0,0)$,

$$f(x,y) = \frac{xy^2}{x^2+y^4}, \qquad g(x,y) = \frac{xy^2}{x^2+y^6}.$$

Prove that f is bounded on $\mathbb{R}^2$, g is unbounded in every neighborhood of $(0,0)$, and f is not continuous at $(0,0)$. However, the restrictions of both f and g to every straight line in $\mathbb{R}^2$ are continuous.

Proof. (1) First, we will show that f is bounded on $\mathbb{R}^2$. For any $(x,y) \in \mathbb{R}^2 \setminus \{(0,0)\}$, we have

$$|f(x,y)| = \frac{|x| \cdot y^2}{x^2+y^4} \leq \frac{|x| \cdot y^2}{2\,|x|\,y^2} = \frac{1}{2}.$$

Hence f is bounded on $\mathbb{R}^2$.

(2) Next, we will show that g is unbounded in every neighborhood of $(0,0)$. For any positive number ε, let $x = \varepsilon^3$ and $y = \varepsilon$, then we have

$$g(x,y) = \frac{\varepsilon^3 \cdot \varepsilon^2}{\varepsilon^6 + \varepsilon^6} = \frac{1}{2\varepsilon},$$

which can be arbitrarily large by choosing sufficiently small ε. Therefore, g is unbounded in every neighborhood of $(0,0)$.

(3) Next, we will show that f is not continuous at $(0,0)$. Let $x = y^2$, then

$$\lim_{y\to 0} f(y^2, y) = \lim_{y\to 0} \frac{y^4}{2y^4} = \frac{1}{2} \neq f(0,0).$$

Then f is not continuous at $(0,0)$.

(4) Finally, we will show that the restrictions of both f and g to every straight line in $\mathbb{R}^2$ are continuous.

- If the line has the form $x = c$, then

  $$f(c,y) = \frac{cy^2}{c^2+y^4}.$$

  If $c = 0$, then $f(c,y) = 0$, which is obviously continous; If $c \neq 0$, then the denominator of f is not zero, and hence $f(c,y)$ is continuous.

- If the line has the form $y = kx + c$ with $c \neq 0$, then

$$f(x, y) = f(x, kx + c) = \frac{x(kx + c)^2}{x^2 + (kx + c)^4}.$$

  Since the denominator is not zero, $f(x, kx + c)$ is continuous.

- If the line has the form $y = kx$, then

$$f(x, y) = f(x, kx) = \frac{k^2x^3}{x^2 + k^4x^4} = \frac{k^2}{k^4x + 1/x} \text{ (for } x \neq 0).$$

  Then $\lim_{x\to 0} f(x, kx) = 0 = f(0, 0)$, and hence $f(x, kx)$ is continuous.

The same argument applies to g. Therefore, the restrictions of both f and g to every straight line in $\mathbb{R}^2$ are continuous. □

Problem 8 (4-14). Let $I = [0, 1]$ be the closed unit interval. Suppose f is a continuous map from I into I. Prove that there exists at least one $x \in I$ such that

$$f(x) = x.$$

Proof. Define

$$A = \{x \in [0, 1] : f(x) \geq x\}.$$

Since $f(0) = 0 \geq 0$, we have $0 \in A$, and therefore $A \neq \varnothing$, and hence $\xi := \sup A$ is a finite number. Moreover, we have $\xi \in [0, 1]$ since $A \subset [0, 1]$.

We will prove that $f(\xi) = \xi$. Denote $g(x) = f(x) - x$, then g is continuous on $[0, 1]$.

If $f(\xi) > \xi$, then $g(\xi) > 0$. Then, there exists $\delta > 0$, such that for any $x \in (\xi - \delta, \xi + \delta)$,

$$|g(x) - g(\xi)| < \frac{g(\xi)}{2} \implies g(x) > \frac{g(\xi)}{2} > 0.$$

Take $x_1 \in (\xi, \xi + \delta)$, we have $f(x_1) > x_1$ and consequently $x_1 \in A$, which contradicts the definition of ξ.

If $f(\xi) < \xi$, then $g(\xi) < 0$. Then, there exists $\delta > 0$, such that for any $x \in (\xi - \delta, \xi + \delta)$,

$$|g(x) - g(\xi)| < \frac{-g(\xi)}{2} \implies g(x) < \frac{g(\xi)}{2} < 0.$$

Take $x_2 \in (\xi - \delta, \xi)$, we have $f(x) < x$ for any $x \in (x_2, \xi)$, and therefore (combining with the definition of ξ) for any $x \in (x_2, 1]$. Then, x_2 is an upper bound less than ξ. This contradicts the definition of ξ.

Therefore, $f(\xi) = \xi$, and there exists at least one $x \in I$ such that $f(x) = x$. □